

Frailty Stratification for the 65+ Population

Data Science Task — Submission (2 pages)

Author: . Approach answers the three required areas; depth over polish.

1. Problem Framing

Objective. Identify residents aged 65+ at risk of *becoming* frail early enough to intervene — to cut avoidable emergency admissions, slow escalation into costly long-term care, and protect independence. The value is in catching the **rising-risk middle**, not just confirming who is already frail. Output: a **Low / Medium / High** risk band + the top reasons per person, refreshed monthly, delivered as prioritised work-lists to community teams and a dashboard for commissioners.

Solution framework / architecture (feature engineering → model training):

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Sources → Data Quality & person-matching → Point-in-time Feature Store
→ Label definition → Train/validate (temporal split) → Calibrate
→ Monthly batch scoring (band + SHAP reasons) → Monitoring (drift/fairness) → Retrain
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Governance (Caldicott Guardian / DPO / DPIA) and **intervention capacity** sit on the critical path — there is no point finding people we cannot help.

Data I would ideally use: - **GP / primary care** — long-term conditions, polypharmacy, consultation frequency *and trend* (basis of the eFI; high value). - **Hospital / SUS** — emergency admissions, A&E, **falls**, length of stay (strong signal *and* the cost outcome we want to reduce). - **Adult social care** — assessments, reablement, care-package hours/escalation (high value, but use carefully — leakage risk). - **Demographics** — age, sex, **deprivation (IMD)**, ethnicity (drivers *and* essential for fairness testing). - **Unstructured case notes** — falls, confusion, "not coping", carer strain, isolation (surfaces deterioration before it is coded). - **Housing / isolation / telecare** — living alone, unsuitable housing, lifeline alarms (isolation proxies).

How frailty would be defined / labelled. No single ground truth, so I frame it as **predicting a future adverse event** — *emergency admission, fall, or care escalation within 12 months* — and use the **electronic Frailty Index (eFI)** and Clinical Frailty Scale as **corroborating features and validation anchors**, not the sole target. *Reasoning:* a predictive label tied to the cost outcome is actionable and measurable; labelling purely on care escalation would just relearn the council's past decisions and their biases. Definition to be clinically signed off and kept configurable.

2. Modelling Approach

How I would explore and understand the data (EDA): - **Missingness** — quantify and treat as informative ("no GP record" ≠ healthy) using missingness-indicator flags; check whether blind spots correlate with deprivation. -

Outliers — implausible ages, negative length-of-stay, duplicate persons (ties back to data-quality/person-matching). - **Class imbalance** — high-frailty is rare (single-digit %); this drives metric and sampling choices. -

Correlation / redundancy — e.g. polypharmacy ↔ condition count. - **Segmentation** — by age band, deprivation decile, condition cluster, geography, to confirm the model works *across* groups.

Step-by-step plan to build the model: 1. **Assemble & match** data to a single resident record (point-in-time correct — only data available *before* the prediction date, to avoid leakage). 2. **Define the label** (future adverse event within 12 months); agree clinically. 3. **Engineer features** (see key features below); add missingness flags. 4. **Split temporally** — train on an earlier period, validate/test on a later one (mirrors live use). 5. **Baseline first** — regularised **Logistic Regression** (transparent benchmark). 6. **Primary model** — **LightGBM / XGBoost**; tune with cross-validation; handle imbalance via `class_weight/scale_pos_weight`. 7. **Calibrate** probabilities (Platt/isotonic) so the Low/Med/High bands are trustworthy. 8. **Set thresholds** to the team's real intervention capacity; produce bands + per-person SHAP reasons. 9. **Validate fairness** across deprivation/ethnicity/sex/age before release.

Key features I would consider: count of long-term conditions & multimorbidity; **polypharmacy** ($\geq 5/\geq 10$ meds, anticholinergic burden); emergency admissions / A&E / **falls** in 6–12m; GP consultation frequency & trend; care-package hours & escalation; age & **deprivation**; living-alone / isolation indicators; **NLP-derived flags** (falls, confusion, "not coping", carer strain) from case notes.

Model types and why: - **Primary** — **LightGBM / XGBoost**: best accuracy on messy, missing, mixed tabular health data; handles non-linearities and missing values natively; **SHAP** gives per-resident explanations. -

Secondary — **Logistic Regression**: transparent odds ratios a GP/commissioner trusts, trivial to govern and

deploy. *If the simple model is close, ship it — adoption beats a leaderboard.*

Evaluation metrics (imbalanced cohort, so accuracy misleads): - **PR-AUC** — primary; robust under imbalance, reflects the rare high-risk class. - **Recall @ the intervention-capacity threshold** — "of the truly at-risk, how many did we catch within the N visits we can actually do?" — the metric that matters most here. - **Precision @ that threshold** — controls alert fatigue / wasted visits. - (Plus **calibration** / **Brier** so the bands mean what they say.) Ultimately, success is validated on **real admissions avoided**, not AUC.

3. Monitoring and Fine-Tuning

Tracking performance after deployment: - **Data drift** — input distributions shift (e.g. a GP coding-system change): monitor via PSI/KS tests on key features and missingness rates. - **Concept drift** — the feature→frailty relationship changes (e.g. post-pandemic care patterns): track rolling performance vs realised outcomes and calibration drift. - **Performance** — back-test PR-AUC / recall / precision once outcomes mature; control charts with alerting.

When to fine-tune / retrain: on a **schedule** (e.g. annually) **and** on **trigger** (drift breach, performance below threshold, or a data-source/policy change). Always temporal-validate and require human sign-off before promoting — never auto-promote a model that decides about vulnerable people.

Keeping the model fair & explainable: use **SHAP** (per-resident and global) alongside the logistic-regression odds ratios as a plain-English benchmark, and **bias-test recall/precision/calibration across deprivation, ethnicity, sex and age** to catch systematic under-flagging of under-served groups. The model **prioritises; professionals decide** (human-in-the-loop). To a non-technical client: *"This is the 200 residents most likely to face a crisis next year, the main reason for each, checked to work fairly across deprived and minority groups — a prioritisation aid, not an automated decision."*